Sean Richards

Research Fellow · Department of Physics

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Education

Doctor of Philosophy (PhD) — Auckland, New Zealand

The University of Auckland — September 2021 - August 2025

- Supervisors: Prof. Jan (JJ) Eldridge & Assoc. Prof. Nicholas Rattenbury
- Thesis: *The Next Generation: Refining the Standard Population Synthesis Model to Better Predict Stellar Population Properties*

Master of Science (MSc) — Auckland, New Zealand

The University of Auckland — March 2020 - March 2021

- Supervisor: Prof. Jan (JJ) Eldridge
- Thesis: *The Missing Mergers: Modelling the observed binary neutron-star population and their mergers*
- Awarded with Honours (First Class)

Postgraduate Diploma in Science (PGDipSci) — Auckland, New Zealand

The University of Auckland — March 2019 - February 2020

- Supervisor: Assoc. Prof. Nicholas Rattenbury
- Dissertation: *The University of Auckland Ground Station One*
- Technical report: *The University of Auckland Ground Station One: System Overview*
- Technical report: *Ground Segment Support for APSS-I* (contains non-public information, and cannot be distributed)

Bachelor of Science (BSc) — Auckland, New Zealand

The University of Auckland — February 2016 - November 2018

- Majored in Physics and Computer Science

Professional Experience

2025-2026 **Technician**, The University of Auckland

Awards, Fellowships, & Grants

2026-2028 Research Fellow
The University of Auckland

2021-2025 University of Auckland Doctoral Scholarship
The University of Auckland

2019 First in Course Award (PHYSICS 746: Quantum Field Theory)
The University of Auckland

Presentations

** presenting author;*

Invited Talks

Richards, S, 2026. *The Next Generation: Refining the Standard Population Synthesis Model to Better Predict Stellar Population Properties*. Colloquium presentation: The University of Auckland Department of Physics, Auckland, New Zealand.

Richards, S, Burn, L. 2024. *Beyond the Bias: Student insights on equity from Auckland to Exeter.* Colloquium presentation: The University of Auckland Department of Physics, Auckland, New Zealand.

Contributed Presentations

Richards, S, 2026. *Binary stars through cosmic time.* Oral presentation: The 3rd Faculty of Science Pacific Research Symposium, Auckland, New Zealand.

Richards, S*, Eldridge, J, Ghodla, S, Briel, M. 2026. *The evolution of accretor stars in binary systems due to accretion of increasingly helium-rich material.* Oral presentation: National Astronomy Meeting 2026, Birmingham, United Kingdom.

Richards, S, 2024. *A fate written in stardust: Probing the interiors of interacting binary stars.* Oral presentation: The 2nd Faculty of Science Pacific Research Symposium, Auckland, New Zealand.

Richards, S*, Eldridge, J, Briel, M, Ghodla, S. 2024. *The variable evolution of accretor stars in binary systems due to accretion of increasingly helium-rich material.* Oral presentation: Annual Scientific Meeting of the Astronomical Society of Australia (online).

Richards, S*. 2022. *New constraints on the Bray conservation-of-momentum natal kick model from multiple distinct observations.* Oral presentation: Supernovae in the Gravitational Wave Detection Era, Melbourne, Victoria, Australia.

Teaching Experience

2026	PHYSICS 753 Course Coordinator	The University of Auckland
2023-2025	PHYSICS 102 Tutor	The University of Auckland
2021-2023	Tuākana Programme Tutor	The University of Auckland
2018-2025	ASTRO 100 / PHYSICS 107 Graduate Teaching Assistant	The University of Auckland

Research Experience

University of Auckland - Department of Physics — Auckland, NZ
Supervisor: Assoc. Prof. Nicholas Rattenbury — June 2023 - December 2023
- Research Assistant: University of Auckland Ground Station One

University of Auckland - Department of Physics — Auckland, NZ
Supervisor: Prof. Jan (JJ) Eldridge — April 2021 - September 2021
- Research Assistant

University of Auckland - Department of Physics — Auckland, NZ
Supervisor: Assoc. Prof. Nicholas Rattenbury — December 2019 - February 2020
- Summer Research Scholar: “Ground Segment Support for APSS-I”
- This report is non-distributable due to a non-disclosure agreement

Outreach & Professional Development

Academic Citizenship and Service

2026- Faculty of Science Equity Committee — The University of Auckland
Member

2025 Department of Physics Equity Committee — The University of Auckland
Chair

2023-2026 Department of Physics Equity Committee — The University of Auckland
Member

Professional Memberships

2026- Royal Astronomical Society — United Kingdom
Fellow

Outreach Activities

2021-2025 Manawa Mai Open Day — The University of Auckland
Demonstrator

2021-2025 Future Me — The University of Auckland
Demonstrator

2021-2025 Incredible Science — The University of Auckland
Demonstrator

Societies

2021 Post-Graduate Students Association — The University of Auckland
Welfare Vice President

2021 Physics Association of the University of Auckland — The University of Auckland
Vice President

2020 Physics Association of the University of Auckland — The University of Auckland
Treasurer

2018 Physics Association of the University of Auckland — The University of Auckland
Committee Member

Professional Development

First Aid Certificate, September 2025 (expiry September 2027)

Publications

Published

Richards, S., Eldridge, J., Briel, M., Stevance, H., Willcox, R. 2023. *New constraints on the Bray conservation-of-momentum natal kick model from multiple distinct observations*. Monthly Notices of the Royal Astronomical Society.

Stevance, H., Ghodla, S., **Richards, S.**, Eldridge, J., Briel, M., Tang, P. 2023. *VFTS 243 as predicted by BPASS models*. Monthly Notices of the Royal Astronomical Society.

Richards S., Eldridge J., Briel M., Ghodla S. 2025. *The evolution of accretor stars in binary systems due to accretion of increasingly helium-rich material*. Monthly Notices of the Royal Astronomical Society.

In Review

In Prep

Mandhai S., Breton R., Kay S., **Richards, S.** 2026. *The orbital evolution of NSNS in Milky Way like galaxies*.

Richards, S., Eldridge, J., Bray, J., Gleave, J., Steed, M. 2026. *The Aotearoa STARS code*.

Richards, S. 2026. *Takahe: Compact object inspirals with BPASS*

Richards, S. 2026. *Kaitiaki: Watching over STARS*